

LOGICCRAFT SOFTWARE SOLUTIONS

AI with Cybersecurity Internship Programme • 2026

INTERNSHIP PROJECT REPORT

URL Safety Checker

Submitted By	:	Fathima Abdul Jabbar
Intern ID	:	ENR-7C1974
Project Type	:	Mini Project
Internship Domain	:	AI with Cybersecurity
Project Guide	:	Mr. Jithin Krishnan P G
Duration	:	2 months
Submission Date	:	20-05-2026

© 2026 LogicCraft Software Solutions. All rights reserved.

CERTIFICATE OF COMPLETION

This is to certify that

Fathima Abdul Jabbar

has successfully completed the AI with Cybersecurity Internship
project titled

"URL Safety Checker "

under the guidance of LogicCraft Software Solutions.

Project Guide

Signature & Date

Authorised Signatory

Signature & Date

DECLARATION

I hereby declare that the project titled

"URL Safety Checker "

is the original work carried out by her during the AI with Cybersecurity Internship at LogicCraft Software Solutions. The project was developed for learning and internship purposes under the guidance of the company mentor. All external sources and references have been duly acknowledged.

Fathima Abdul Jabbar

Intern ID: ENR-7C1974

Signature & Date

ACKNOWLEDGEMENT

I express my sincere gratitude to LogicCraft Software Solutions for providing the opportunity to undergo internship training in AI with Cybersecurity . This experience has been invaluable in bridging the gap between academic knowledge and real-world application.

I express my sincere thanks to her project guide, Mr. Jithin Krishnan P G, for patient guidance, continuous support, and constructive feedback throughout the internship period. His expertise and mentorship played a pivotal role in the successful completion of this project.

I also extends my gratitude to all team members and mentors at LogicCraft Software Solutions for creating a collaborative and encouraging learning environment.

Finally, I thanks her family and friends for their constant encouragement and motivation throughout this journey.

ABSTRACT

The URL Safety Checker is a web-based cybersecurity application developed using Python and Flask to identify whether a URL is safe, suspicious, or invalid. The primary objective of this project is to create awareness about potentially harmful websites and phishing links by providing a simple URL analysis system. The application accepts a URL input from the user and performs validation to determine whether the entered text is a properly formatted URL. It further analyzes the URL using predefined suspicious keywords commonly associated with phishing and malicious websites, such as “login”, “verify”, “secure”, “free”, and “prize”. Based on the analysis, the system classifies the URL as Safe URL, Unsafe URL Detected, or Invalid or Unsafe URL. The project was implemented using Python Flask for backend development and HTML/CSS for the frontend interface. The application provides a user-friendly interface with color-based result indicators for improved readability and user experience. This project demonstrates the practical application of basic cybersecurity concepts, URL validation techniques, and web application development in creating a simple and effective security-focused solution.

1. Introduction

AI with Cybersecurity is a critical discipline in today's digital-first world, safeguarding systems, networks, and user data from unauthorised access and malicious attacks.

The URL Safety Checker project was built using the following technologies:

- Python 3.14.5
- Flask
- HTML
- CSS
- Visual Studio Code
- Render

This project was completed as part of a 2 months AI with Cybersecurity Internship at LogicCraft Software Solutions by Fathima Abdul Jabbar.

2. Company Profile

LogicCraft Software Solutions is a technology training and software development organisation with a strong focus on equipping aspiring developers and security professionals with industry-relevant, hands-on skills.

2.1 About LogicCraft

Founded with the mission to bridge the gap between academia and industry, LogicCraft specialises in delivering practical internship programmes across web development, cybersecurity, data analytics, and Python programming.

2.2 Services Offered

- Web Application Development
 - AI in Cybersecurity
 - Automation & Multi-Agent Systems
 - NLP & Conversational AI
 - AI-Powered Data Science
 - Advanced Computer Vision
 - AI with Cybersecurity Training & Awareness
 - Python Programming Courses
 - Structured Internship Programmes
-

- Software Testing & Quality Assurance

3. Project Objectives

The URL Safety Checker was designed with the following key objectives:

- To helps users identify suspicious and potentially unsafe URLs..
- To provides basic awareness about phishing and malicious websites..
- To validates whether the entered text is a proper URL..
- To simple and user-friendly interface for easy interaction..
- To displays results clearly using color-based indicators..

4. System Requirements

4.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	Minimum 4 GB
Storage	Minimum 500 MB free space
System Type	64-bit Operating System
Keyboard and Mouse	—

4.2 Software Requirements

Software	Details
Operating System	Windows 10 or above
Programming Language	Python 3.14.5
Framework	Flask
Code Editor	Visual Studio Code
Web Browser	Google Chrome / Microsoft Edge

5. Modules of the Project

5.1 URL Validation Module

The URL Validation Module is an important part of the URL Safety Checker application. Its main function is to verify whether the text entered by the user is a properly formatted URL before the system performs further analysis. This helps the application avoid incorrect results and ensures that only valid website links are processed. When a user enters a URL, the module checks for essential elements such as the protocol (`http://` or `https://`) and a valid domain name. The validation process is implemented using Python's URL parsing functionality. If the entered input does not follow the standard URL format, the system identifies it as an invalid or unsafe URL and informs the user immediately. This module improves the overall accuracy and reliability of the application. For example, random inputs such as `google` or `12345` are rejected because they are not complete URLs, while valid links such as `https://google.com` are accepted and analyzed further. By filtering invalid inputs at an early stage, the module helps the application produce more meaningful and accurate results. In addition to improving system performance, the URL Validation Module also enhances user experience by providing clear and immediate feedback. Overall, this module plays a significant role in maintaining the effectiveness and usability of the URL Safety Checker application while supporting its basic cybersecurity functionality.

8. Advantages

- Helps users identify suspicious and potentially unsafe URLs.
- Provides basic awareness about phishing and malicious websites.
- Validates whether the entered text is a proper URL.
- Simple and user-friendly interface for easy interaction.
- Displays results clearly using color-based indicators.
- Reduces the chances of users accessing unsafe links accidentally.
- Lightweight application with fast response time.
- Easy to develop, maintain, and upgrade with additional security features.
- Demonstrates practical implementation of basic cybersecurity concepts.
- Useful for learning web development using Python Flask and frontend technologies.

9. Limitations

- The application checks URLs using only basic validation methods.
- It does not use real-time online threat detection.
- Some advanced phishing links may not be detected.

- Machine learning techniques are not included in the project.
- Features like SSL verification and blacklist checking are not implemented.

10. Future Enhancements

- Real-time threat detection APIs can be integrated to improve the accuracy of URL analysis and identify malicious websites more effectively.
- Machine learning techniques can be implemented to detect advanced phishing URLs and improve the overall security performance of the application.
- Blacklist checking and domain reputation analysis features can be added to verify whether a website is reported as unsafe or suspicious.
- The user interface can be further enhanced with improved design, animations, and additional user-friendly features for a better experience.
- A database system can be developed to store scanned URLs, maintain analysis history, and generate reports for future reference.

11. Learning Outcomes

- Gained practical knowledge in developing web applications using Python and the Flask framework.
- Understood the basics of URL validation and the identification of suspicious website links using predefined security-related keywords.
- Improved understanding of fundamental cybersecurity concepts such as phishing detection and unsafe URL identification.
- Developed skills in frontend and backend integration using HTML, CSS, and Flask for creating interactive web applications.
- Improved problem-solving and debugging skills while developing and testing the application.

12. Conclusion

The URL Safety Checker project was developed to provide a simple and effective solution for identifying safe, suspicious, and invalid URLs. The application performs URL validation and basic keyword-based analysis to help users recognize potentially unsafe website links. Developed using Python Flask, HTML, and CSS, the project demonstrates the practical implementation of basic cybersecurity concepts along with frontend and backend web development techniques. The system also provides a user-friendly interface with clear result indicators, making it simple and easy to use.

The development of this project helped in gaining practical knowledge in areas such as web application development, URL validation, phishing awareness, and debugging techniques. Although the current version uses basic analysis methods, it provides a strong foundation for future enhancements and advanced security features. Overall, the project serves as an effective beginner-level cybersecurity application that combines simplicity, functionality, and practical learning experience.

13. References

- Python Official Documentation. Available at: <https://www.python.org/>
- Flask Official Documentation. Available at: <https://flask.palletsprojects.com/>
- MDN Web Docs - HTML Documentation. Available at: <https://developer.mozilla.org/>
- W3Schools Web Development Tutorials. Available at: <https://www.w3schools.com/>
- CSS Official Documentation. Available at: <https://developer.mozilla.org/en-US/docs/Web/CSS>
- Render Deployment Documentation. Available at: <https://render.com/docs>

— End of Internship Report —

Fathima Abdul Jabbar | ENR-7C1974 | URL Safety Checker